CSA 1036

THEORY OF COMPUTATION!

CO-02

AT-02

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Task 1 : Mathematical Formulation

① Input alphabet:

$$\Sigma = \{0, 1\}$$

② Examples of strings :-

- 101
- 110
- 001
- 1110

③ Formal language:

L is the set of all binary strings that start and end with same symbol.

④ Types of strings :-

- Finite string : 1010
- Empty string : ε
- Language : set of strings over an alphabet

⑤ Finite Automaton:

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q - Set of states

Σ - input alphabet

δ - Transition function

q_0 - initial state

F - Final states.

6. Role of states :-

- States represent the current position.
- Initial state - starting state ; Final state : indicate acceptance

→ Task 2 : DFA - strings starting and Endings with Same Symbol.

States : Q =

q_0 - initial state

q_1 - starts with 0, ends with 0.

q_2 - starts with 1, ends 1

q_3 - Rejected State

Final states : q_1, q_2

Transition Table :

state	0	1
→ q_0	q_1	q_2
* q_1	q_1	q_3
* q_2	q_3	q_2
q_3	q_1	q_2



Examples :

Input	Result
0	Accepted
11	Accepted
101	Accepted
01	Rejected.

→ Task 3 : NFA - substring 101

States : Q =

- q_0 - initial state
- q_1 - Found 1
- q_2 - Found 10
- q_3 - Found 101 (final state)

Transition table:

State	0	1
→ q_0	q_0	q_0, q_1
q_1	q_2	-
q_2	-	q_3
* q_3	q_3	q_3

Accepted :

101 , 1101 , 1010

Rejected :-

000, 111, 100.

→ Task 4 : NFA to DFA Conversion

Subset states :-

- A -
- B -
- C -
- D -
- E -
- F -

initial state : A

Final state : E, F

Transition Table:

State	0	1
A	A	B
B	C	B
C	D	E
D	A	E
E	C	F
F	A	F

The DFA & NFA recognize the same language: Binary strings containing 101.

→ Task -5 - ε-NFA - pattern 110.

States : q_0, q_1, q_2, q_3

Transitions :

- $q_0 - \varepsilon \to q_1$
- $q_1 - 1 \to q_2$
- $q_2 - 1 \to q_3$
- $q_3 - 0 \to q_4$

For a complete pattern recognition automaton, use q_0 as the scanning state and q_4 as the final state, with loops on 0 and 1 at q_0 & a transition from q_0 to q_1 on 1.

ε - closure :-

- ε - closure (q_0) =
- ε - closure (q_1) =
- ε - closure (q_2) =
- ε - closure (q_3) =

48/50

Examples :-

- 110 - Accepted
- 0110 - Accepted
- 111 - Rejected
- 000 - Rejected.